

Evaluate the short and long-term consequences of the recent US–Iran conflict on the oil market.

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Abstract

This paper evaluates how the US–Iran conflict from February 2026 impacted the global oil market by analysing the dynamics of the Brent–WTI spread. As both benchmarks track light, sweet crude, the spread serves as an ideal analytical instrument; by contrasting landlocked WTI with waterborne Brent, it effectively isolates geopolitical risk from physical crude quality differences. Utilising this framework, I examine whether the conflict induced a temporary, albeit significant, premium or a permanent structural shift in the oil market’s geopolitical landscape. I demonstrate that while the conflict caused a severe deviation in the cointegrating relationship—surpassing prior Middle Eastern supply shocks—no permanent structural break seems to have occurred. Finally, I show that these findings motivate a regime-aware volatility overlay that manages tail risk during crisis vol spikes, reducing maximum drawdown by over 82% relative to a naive mean-reversion baseline; the primary contribution is tail-risk management rather than a direct improvement to the mean-reversion edge.

1 Existence of the relationship

The Brent–WTI spread exhibits a well-documented cointegrating relationship.¹ To verify and exploit this relationship, I investigated daily spot prices between 1987 and 2026, sourced from the U.S. Energy Information Administration (EIA).

Augmented Dickey–Fuller tests confirm that both log-price series are integrated of order one, $I(1)$.² Cointegration then requires a stationary linear combination, so we estimate the Engle–Granger long-run relationship $\log B_t = \alpha + \beta \log W_t + \varepsilon_t$ and test the residual ε_t for stationarity. When $\varepsilon_t \sim I(0)$ the benchmarks are cointegrated. β measures the long-run cross-price elasticity of Brent on WTI, and the disequilibrium ε_t feeds an error-correction model.

The error-correction model is

$$\Delta \log B_t = c + \gamma \varepsilon_{t-1} + \sum_{j=1}^p \phi_j \Delta \log B_{t-j} + \sum_{j=1}^p \psi_j \Delta \log W_{t-j} + u_t, \quad (1)$$

where c is a drift constant and ε_{t-1} the lagged disequilibrium from the long-run relationship, while the ϕ_j and ψ_j capture short-run dynamics through lagged price changes. The disturbance u_t is a zero-mean innovation, assumed serially uncorrelated with finite variance (white noise) — the part of the daily Brent return left unexplained by error correction and short-run momentum. A negative adjustment coefficient γ then implies mean reversion with half-life $H = -\ln 2 / \ln(1 + \gamma)$.

Estimating these relationships over different sub-periods confirms the relationship is largely stable (Table 1). β sits close to unity throughout, and the Johansen trace test confirms exactly one

*Code, data pipeline and figures: <https://github.com/charliecockerell/brent-wti-regime-shift>

¹Caporin, Fontini, and Talebbeydokhti 2019; Luong, Mizrach, and Otsubo 2019.

²Dickey and Fuller 1979; Engle and Granger 1987.

cointegrating vector at the 5% level in each window.³ Cointegration holds at the 5% level in every window, including one that excludes the contested 2010–2015 divergence,⁴ and the half-life stays short.

Table 1: Cointegration estimates across data windows (EIA daily spot, log prices).

Window	n	β [95% CI]	EG cointegration p	Half-life (days)
Full 1987–2026	9734	1.094 [1.086, 1.101]	< 0.001	24.2
Post-2005	5332	1.035 [1.011, 1.058]	0.008	29.6
Post-2010	4081	1.048 [1.027, 1.069]	< 0.001	23.1
Post-2015	2830	1.018 [0.999, 1.037]	< 0.001	10.4
Excl. 2010–2015	8229	1.081 [1.073, 1.089]	< 0.001	6.8

n = number of trading days in each window.

2 Short-term consequences of the US–Iran conflict on the relationship

Despite the strong link established in Section 1, geopolitical events can act as severe supply shocks, with prices often moving on news before any physical disruption materialises.⁵ I therefore ran an event study of five Middle Eastern supply shocks with a dateable catalyst, in chronological order:

- **Aramco** (16 Sep 2019). Strike on Abqaiq–Khurais.
- **Soleimani** (3 Jan 2020). US strike killing the IRGC Quds Force.
- **Houthi** (19 Jan 2024). Red Sea shipping attacks.
- **Epic Fury** (28 Feb 2026). Opening strike of the conflict.
- **Hormuz closure** (4 Mar 2026). Strait of Hormuz disruption.

The first three are historical comparables; the last two are the 2026 conflict. For each event we compute the spread z -score, $z_t = (s_t - \bar{s}_{60})/\hat{\sigma}_{60}$, where s_t is the dollar spread and the mean and standard deviation are estimated over a trailing 60-day window.

Table 2: Event-study metrics across the five supply shocks (EIA daily spot, dollar spread).

Event	Peak z	Days to peak	Pre-vol	Post-vol	Vol ratio
Aramco (Sep 2019)	−0.88	20	1.30	0.76	0.59
Soleimani (Jan 2020) †	−2.95	20	0.66	1.02	1.53
Houthi (Jan 2024)	3.06	9	1.25	1.48	1.18
Epic Fury (28 Feb 2026)	11.63	12	0.77	6.98	9.01
Hormuz closure (4 Mar 2026)	12.50	19	0.89	7.10	7.94

† Soleimani censored at +20 days (avoids the March-2020 COVID crash). Peak $z = \max |\text{deviation}|$ in $[0, +20]$ d; vol ratio = post $\div$ pre realised spread vol.

Despite similarities, earlier shocks barely moved the spread. 2026 was categorically different with post-event volatility $\sim 8\times$ the pre-event baseline, against $\sim 1.2\times$ historically (Figure 2). A

³Johansen 1995.

⁴Caporin, Fontini, and Talebbeydokhti 2019; Gregory and Hansen 1996; Luong, Mizrach, and Otsubo 2019; Zivot and Andrews 1992.

⁵Kilian, Plante, and Richter 2024; Verduzco-Bustos and Zanetti 2026.

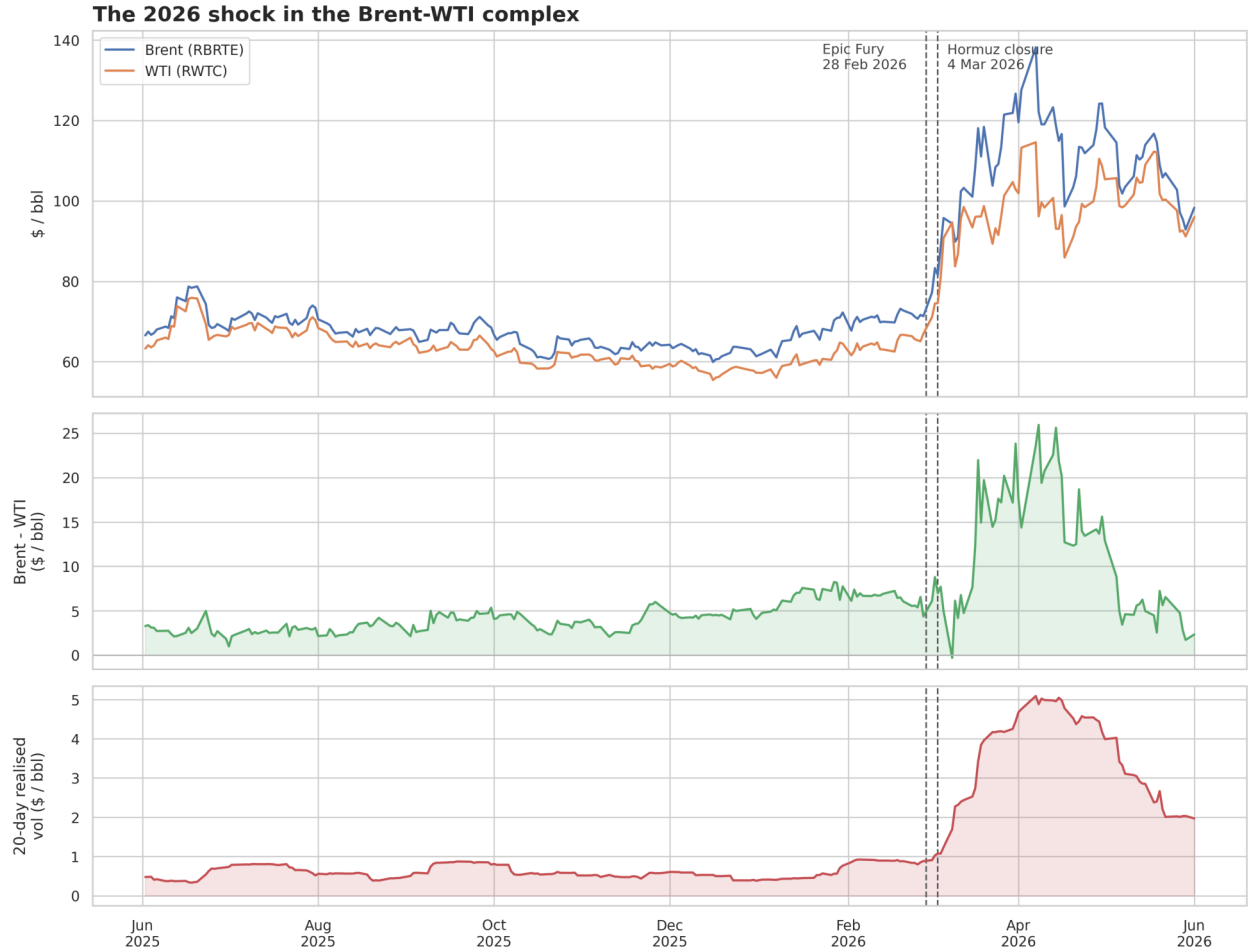


Figure 1: Brent and WTI levels, the dollar spread, and 20-day realised spread volatility, mid-2025 to the data edge; dashed lines mark Epic Fury (28 Feb) and Hormuz (4 Mar).

Levene test confirms that post-event spread-change variance in 2026 is significantly higher than in all three historical shocks combined ($W = 76.7$, $p < 0.001$).

That sharpness is the key short-term consequence: the 2026 dislocation peaked near $+12\sigma$ against a historical bound of $\approx 3\sigma$, a scale never observed in the prior record and qualifying as a genuine tail-risk event rather than a large-but-normal outlier.

3 Long-term consequences: resilience, not rupture

The conflict does not seem to have permanently changed the relationship. Testing gives a post-event β of 1.012, overlapping the pre-event and full-sample β , with cointegration holding in every sub-sample (Table 3). The lasting impact on this relationship seems negligible: the spread dislocated violently, but the cointegration did not break.

This null is not a failure of statistical power. The rolling 2-year β and half-life (Figure 3, Table 3) act as a within-sample positive control: both move violently at the 2011 shale revolution — the only clean structural break in forty years⁶ which Zivot–Andrews and Gregory–Hansen date to December 2010 — with β swinging away from 1.0, and the half-life spiking to 147 days. Through the 2026 conflict the same detector is flat: β holds in 0.91–1.06 and the half-life never exceeds ~ 8 days (on the rolling 2-year window used here; the full-sample 24-day figure in Table 1 is pulled upward by

⁶Caporin, Fontini, and Talebbeydokhti 2019; Gregory and Hansen 1996; Luong, Mizrach, and Otsubo 2019; Zivot and Andrews 1992.

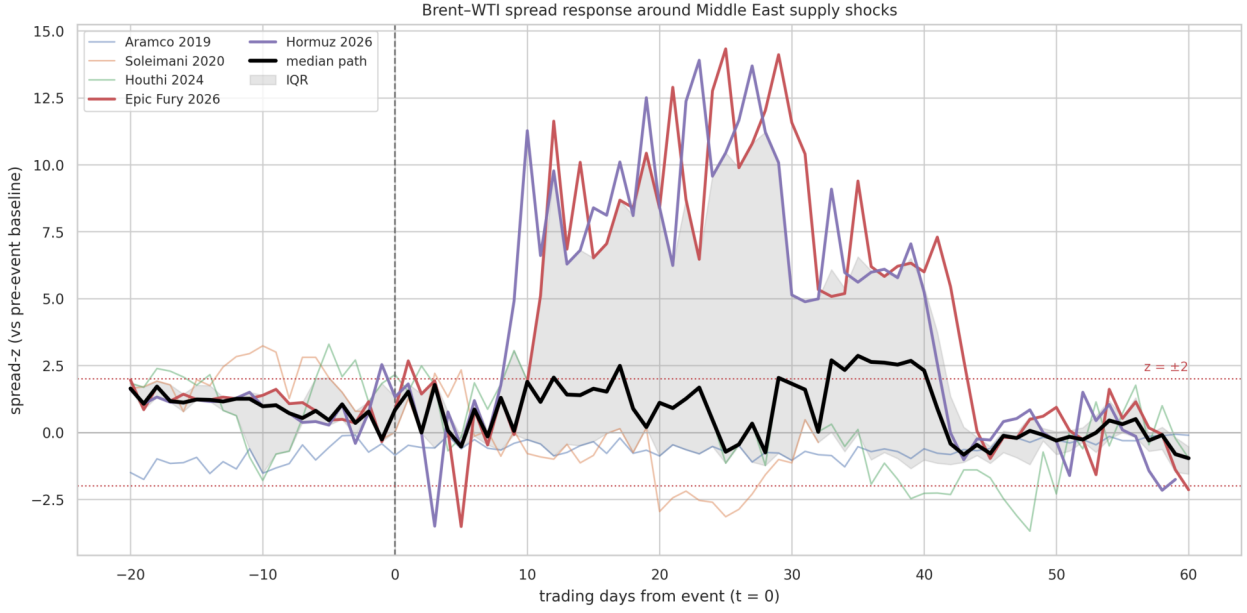


Figure 2: Spread z -score paths around each shock, aligned at $t = 0$, with cross-event median and IQR band. 2026 peaks near $+12\sigma$; historical shocks stay within $\approx 3\sigma$.

the slower-reverting pre-2010 era). The instrument that screams at a genuine structural break is silent here.

This is not ‘no long-term consequence’. The Brent–WTI spread is a relative-value lens — US versus waterborne crude — and is silent on effects that move both legs together. The conflict’s durable footprint lives in those dimensions, not in the US–global crude linkage.

Two caveats remain. The endogenous break tests cannot reach February 2026; a targeted Chow breakpoint test⁷ at the conflict date rejects ($F(2, 9730) = 10.2$, $p < 0.001$), but the break is in the spread’s level and variance—the dislocation—not the cointegrating slope. A Chow predictive-failure test gives $F(62, 9670) = 1.42$ ($p = 0.017$): marginally significant, though the classical Chow F over-rejects when residuals are autocorrelated—as cointegrating residuals typically are—and this value is well below what a genuine structural shift produces (see Table 3, 2011 column). 62 observations also cannot entirely rule out a slow, shale-style repricing — a post-conflict ‘drill, baby, drill’ surge is one channel that could still break the relationship. My reading is ‘no break yet’.

Table 3: The 2011 shale break as a positive control for the 2026 test.

Signal	2011 shale break	2026 conflict
Rolling 2-yr β	1.0 $\rightarrow$ 1.46 $\rightarrow$ 0.10 (violent swing)	0.91–1.06 (post- β 1.012, CI spans unity)
Rolling half-life	spikes to 147 days	≤ 8.4 d (≈ 6.5 d baseline)
Endogenous break (ZA / GH)	dated Dec 2010 — detected	test window cannot reach 2026
Chow test at date	$F(2, 9730) = 2858$, $p < 0.001$	$F(2, 9730) = 10.2$, $p < 0.001$: level & variance, β unchanged
Predictive-failure test	$F(3851, 5881) = 3.86$, $p < 0.001$	$F(62, 9670) = 1.42$, $p = 0.017$ (marginal; inflated by autocorr.)
Spread z at data edge	—	≈ -1.35 (reverted)

⁷The Chow breakpoint test partitions the sample at the candidate date and asks whether pre- and post-break coefficients are jointly equal (F -test on pooled vs. segmented RSS); the predictive-failure variant asks whether post-break observations are consistent with the pre-break model, and is better suited when the post-break window is short. (Chow 1960)



Figure 3: Rolling 2-year cointegrating β (top, with 2024–26 zoom inset) and mean-reversion half-life (bottom). The 2011 shale break is unmistakable — β swings violently and the half-life spikes to ~ 150 days — whereas 2026 leaves both flat: a transient dislocation, not a structural break.

4 Trading implications

A naive mean-reversion rule — parameters fixed a priori — earns an out-of-sample (2019–2026) Sharpe of 1.02 across 63 trades (83% hit rate, +\$88/bbl); this Sharpe is significant under a 20-day block bootstrap (95% CI [0.33, 1.65]) and a z -score permutation test ($p < 0.001$, 2,000 shuffles), so the edge is not a random-walk artefact. The 2σ divergence entry, next-day execution and net-of-cost reporting follow a standard relative-value convention,⁸ with the z -score normalisation as in established statistical-arbitrage practice.⁹

But the strategy’s worst drawdown, $-\$26/\text{bbl}$, is the 2026 crisis itself. The mechanism is structural: the rule shorts the spread at $+2\sigma$ and is stopped out at $+3.5\sigma$ as the spread marches to $+12\sigma$ — picking up pennies in front of the steamroller.

The two findings combine into one rule: the cointegration is the anchor; the regime decides deployment. Because the relationship is resilient the reversion will come—just not on a timescale a fixed stop survives. When 20-day realised vol exceeds its trailing-year 80th percentile,¹⁰ we suppress new entries and apply inverse-volatility sizing; calm otherwise. Together this reduces maximum drawdown by 82% ($-\$26 \rightarrow -\$4.6/\text{bbl}$) while lifting OOS Sharpe from 1.02 to 1.42 (Table 4, Figure 4). This observable, ex-ante volatility gate is a deliberately transparent alternative to latent hidden-Markov approaches;¹¹ rather than assuming a regime-switching cointegration vector, we test it directly and find it intact.

⁸Gatev, Goetzmann, and Rouwenhorst 2006.

⁹Avellaneda and Lee 2010.

¹⁰A grid search over the 60th–95th percentile range yields gated Sharpe of 1.18–1.34 throughout, with no threshold sharply dominating; the 80th was fixed before OOS evaluation.

¹¹Fanelli, Fontana, and Rotondi 2023.

Table 4: Backtest summary — naive rule and regime-adaptive overlay (OOS 2019–2026; net of \$0.10/bbl costs).

Strategy	PnL (\$/bbl)	Sharpe (gross)	Sharpe (net)	Max DD (\$/bbl)
Naive	88.4	1.02	0.88	−26.0
Regime-gated	70.1	1.32	1.16	−5.6
Gated + vol-scaled	62.2	1.42	1.25	−4.6
Always-long benchmark	−3.1	−0.02	—	−24.2

Naive: 63 trades, 83% hit rate; train (1987–2018) Sharpe 0.99 — no out-of-sample decay. Always-long: holds one unit long the spread (long Brent / short WTI) throughout the OOS period; no costs deducted. Net-of-cost robustness: the regime-gated Sharpe declines smoothly from 1.32 (gross) to 1.16 (\$0.10/bbl) to 0.99 (\$0.20/bbl), and the drawdown advantage over the naive strategy persists across the full cost range tested.

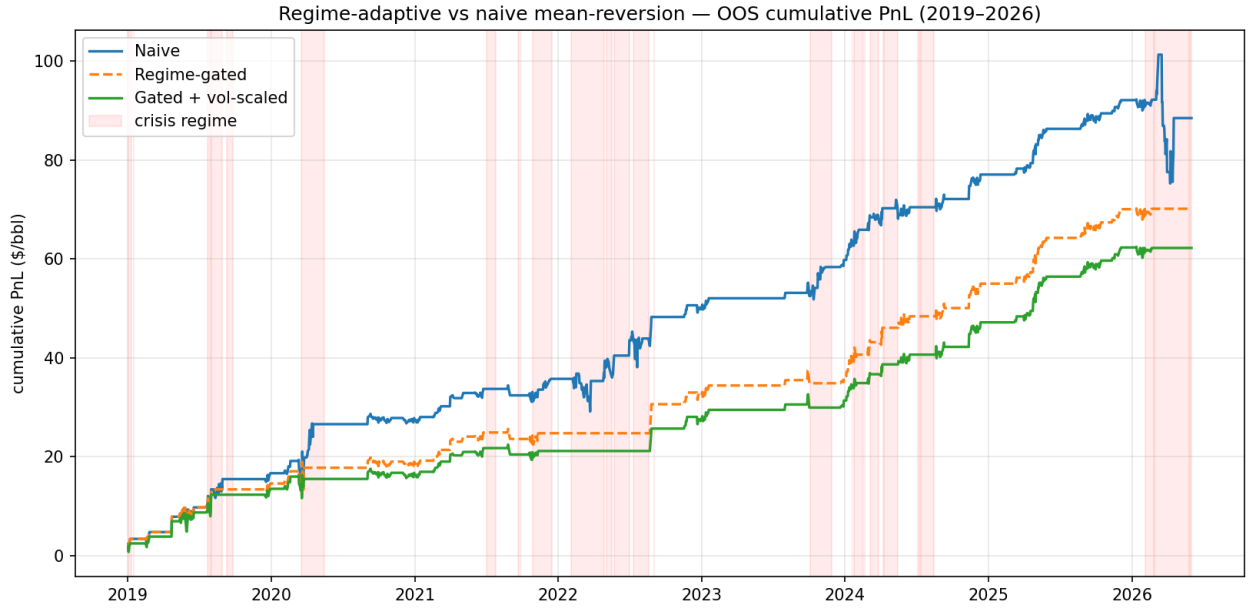


Figure 4: Regime-adaptive vs naive OOS cumulative PnL; shaded bands mark the crisis regime ($\sim 25\%$ of days). Gating and vol-scaling give up some absolute PnL but flatten exposure through the shock.

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